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Electronic hi-hat and operating method of electronic hi-hat

Abstract

The disclosure provides an electronic hi-hat and an operating method of the electronic hi-hat that can improve the feeling at the time of hitting. Since an edge portion of a bottom frame is formed in a flat plate shape, the weight of an outer edge portion of the bottom frame can be reduced. As a result, even if a hit top cymbal comes into contact with the bottom frame, the weight of the outer edge portion of the bottom frame is less likely to hinder the swinging (displacement caused by the hit) of the top cymbal. Therefore, the feeling at the time of hitting the top cymbal can be lightened, so that the feeling at the time of hitting can be improved.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefits of Japanese application no. 2021-057602, filed on Mar. 30, 2021. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to an electronic hi-hat and an operating method of the electronic hi-hat, and more particularly relates to an electronic hi-hat and an operating method of the electronic hi-hat that can improve the feeling at the time of hitting.

Description of Related Art

(3) An electronic hi-hat is known, which includes a bottom cymbal that is swingably attached to a first rod, and a top cymbal that is swingably attached to a second rod which moves up and down with respect to the first rod. This type of electronic hi-hat is played by moving the second rod up and down to bring the top cymbal into contact with the bottom cymbal or hitting the top cymbal. During such a performance, the top cymbal and the bottom cymbal may swing in a state of being in contact with each other.

(4) For example, Patent Document 1 describes a technique of providing a sliding tube **140** or a sliding film **212** at a contact portion between a top cymbal pad portion **100** (top cymbal) and a bottom cymbal pad portion **200** (bottom cymbal). According to this technique, when the top cymbal swings while being in contact with the bottom cymbal, the sliding of the sliding tube **140** and the sliding film **212** can smooth the swinging of each cymbal. Therefore, the feeling at the time of hitting can be close to that of hitting an acoustic hi-hat cymbal.

RELATED ART

Patent Document

(5) [Patent Document 1] Japanese Laid-Open No. 2005-208555 (for example, paragraphs 0099, 0128, 0188, FIG. 5, and FIG. 14).

Problems to be Solved

(6) However, in the above-mentioned conventional technique, in order to secure strength (suppress damage), besides that the outer edge portion of the bottom cymbal (bottom frame) is formed to be relatively thick, the sliding film **212** is attached to the thick portion via a metal plate **210**. Therefore, the weight of the outer edge side (portion in contact with the top cymbal) of the bottom cymbal increases. Due to the increase in the weight of the bottom cymbal, the feeling at the time of hitting the top cymbal may become heavy, which causes a problem that the feeling at the time of hitting cannot be sufficiently improved.

SUMMARY

(7) The disclosure provides an electronic hi-hat and an operating method of the electronic hi-hat that can improve the feeling at the time of hitting.

Means for Solving the Problems

(8) An electronic hi-hat according to the disclosure includes a bottom cymbal that is swingably attached to a first rod; and a top cymbal that is arranged on an upper side of the bottom cymbal and is swingably attached to a second rod which is relatively displaced up and down with respect to the first rod. The bottom cymbal includes a first frame that forms a skeleton of the bottom cymbal, and the first frame includes a contact portion that has a flat plate shape and comes into contact with the top cymbal when the top cymbal is displaced.

(9) An operating method according to the disclosure is provided for an electronic hi-hat, which includes a bottom cymbal that is swingably attached to a first rod, and a top cymbal that is arranged on an upper side of the bottom cymbal and is swingably attached to a second rod which is relatively displaced up and down with respect to the first rod. The bottom cymbal includes a first frame that forms a skeleton of the bottom cymbal. The first frame includes a contact portion that has a flat plate shape. The top cymbal is brought into contact with the contact portion when the top cymbal is displaced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view of an electronic hi-hat according to an embodiment.

(2) (a) of FIG. 2 is a partially enlarged cross-sectional view of the electronic hi-hat in which the IIa portion of FIG. 1 is enlarged, and (b) of FIG. 2 is a partially enlarged cross-sectional view of the electronic hi-hat showing a state where a top cymbal is closed from the state of (a) of FIG. 2.

(3) (a) of FIG. 3 is a partially enlarged cross-sectional view of the electronic hi-hat showing a state where the top cymbal is hit from the state of (b) of FIG. 2, and (b) of FIG. 3 is a partially enlarged cross-sectional view of the electronic hi-hat showing a state where a bottom cymbal and the top cymbal are displaced beyond an initial position from the state of (a) of FIG. 3.

DESCRIPTION OF THE EMBODIMENTS

(4) Hereinafter, exemplary embodiments will be described with reference to the accompanying drawings. First, the overall configuration of an electronic hi-hat 1 will be described with reference to FIG. 1. FIG. 1 is a cross-sectional view of the electronic hi-hat 1 according to an embodiment. FIG. 1 shows a cross section cut along a plane including axes of a bottom cymbal 2 and a top cymbal 3. Further, in order to simplify the drawing, FIG. 1 omits a part of the configuration of the electronic hi-hat 1 and only shows the main parts.

(5) As shown in FIG. 1, the electronic hi-hat 1 includes the bottom cymbal 2 and the top cymbal 3 that moves up and down on the upper side of the bottom cymbal 2, and is configured as an electronic percussion instrument that imitates an acoustic hi-hat cymbal.

(6) The bottom cymbal 2 includes a bottom frame 20 that forms the skeleton thereof, and the bottom frame 20 is supported by a shaft 5 via a bottom support member 4. In FIG. 1, hatching is provided by omitting the detailed cross-sectional structure of the bottom support member 4. The shaft 5 is formed in a cylindrical shape and is configured to be self-supporting by a stand (not shown), and the substantially cylindrical bottom support member 4 is fixed to the upper end of the shaft 5.

(7) The bottom support member 4 is composed of a large diameter portion 40 fixed to the shaft 5, and a small diameter portion 41 formed on the upper surface of the large diameter portion 40. The diameter of the small diameter portion 41 is formed smaller than that of the large diameter portion 40. A through hole 20a having a diameter smaller than that of the large diameter portion 40 of the bottom support member 4 is formed in the center of the bottom frame 20, and the small diameter portion 41 is inserted into the through hole 20a. As a result, the bottom frame 20 is swingably supported on the upper surface of the large diameter portion 40 of the bottom support member 4.

(8) A rod **6** for moving the top cymbal **3** up and down is inserted into the bottom support member **4** and the shaft **5**. The rod **6** moves up and down on the inner peripheral side of the bottom support member **4** and the shaft **5** by operating a step-on pedal (not shown).

(9) A tubular member **7** having a tubular shape is attached to the upper end side of the rod **6**, and a top support member **8** for swingably supporting the top cymbal **3** is fixed to the lower end side of the tubular member **7**. The top support member **8** is formed in a substantially cylindrical shape, and by inserting the tubular member **7** into a through hole **31b** formed in a head portion **31a** of the top cymbal **3**, which will be described later, the top cymbal **3** is hooked on the top support member **8**.

(10) The upper end of the top support member **8** is formed in a mountain shape, and a valley-shaped groove to be hooked on the mountain-shaped upper end of the top support member **8** is formed on the lower surface of the top cymbal **3** (head portion **31a**). As a result, the top cymbal **3** is swingably supported by the top support member **8**.

(11) A washer **9**, a pair of nuts **10**, and a clutch **11** are sequentially mounted on the tubular member **7** so as to overlap with the top cymbal **3**. A male screw is formed on the outer peripheral surface of the tubular member **7**, and by tightening the pair of nuts **10** to press the washer **9** of an elastic member, the top cymbal **3** is swingably mounted to the tubular member **7**.

(12) The clutch **11** is fixed to the upper end of the tubular member **7**, and while the tubular member **7** (top cymbal **3**) is fixed to the rod **6** by tightening a wing bolt **12** from a side surface of the clutch **11** toward the rod **6**, the relative position of the tubular member **7** (top cymbal **3**) with respect to the rod **6** is adjusted by loosening the wing bolt **12**. The rod **6** with the tubular member **7** fixed by the clutch **11** is moved up and down by depressing the pedal (not shown), and in conjunction with this, the top cymbal **3** is displaced up and down (see FIG. 2).

(13) A top frame **30** constituting the skeleton of the top cymbal **3** is composed of a first top frame **31** supported by the top support member **8**, and a second top frame **32** to which the first top frame **31** is fixed and which has an upper surface configured as a striking surface.

(14) The first top frame **31** includes a substantially columnar head portion **31a** formed in the central portion thereof. The through hole **31b** that penetrates vertically is formed in the head portion **31a**, and as described above, the tubular member **7** is inserted into the through hole **31b**.

(15) From the lower end of the head portion **31a**, a flange portion **31c** projects outward in the radial direction in a flange shape. From the outer edge of the flange portion **31c**, a cover portion **31d** hangs downward and projects outward in the radial direction, and the cover portion **31d** covers the lower surface of the second top frame **32**.

(16) The second top frame **32** is formed in a disk shape having a through hole **32a** in the center, and the head portion **31a** of the first top frame **31** is inserted into the through hole **32a** of the second top frame **32**. In this inserted state, the flange portion **31c** of the first top frame **31** is screwed to the inner edge portion of the second top frame **32** and the outer edge portion of the cover portion **31d** is screwed to the lower surface of the second top frame **32**, so as to form the top frame **30** in which the first top frame **31** and the second top frame **32** are integrated.

(17) Electronic components such as an impact sensor **S1** are housed in a space between the cover portion **31d** of the first top frame **31** and the second top frame **32**. The impact sensor **S1** is a piezoelectric element attached to the lower surface of the second top frame **32**, and vibration that occurs at the time of hitting the second top frame **32** is detected by the impact sensor **S1**. A hit detected by the impact sensor **S1** or a pressure-sensitive sensor **S2** (see FIG. 2) described later is converted into an electric signal and output to a sound source device (not shown). As a result, a musical tone corresponding to the striking position on the top cymbal **3** is generated.

(18) The second top frame **32** is formed in a shape that gently slopes downward from the inner edge side to the outer edge side, and the upper surface of the second top frame **32** is covered by a disk-shaped cover **33** that follows this shape. A through hole **33a** is formed in the center of the cover **33**, and the head portion **31a** of the first top frame **31** is inserted into the through hole **33a**.

(19) Since the material (rubber in this embodiment) of the cover **33** is softer than the material (glass

fiber reinforced resin in this embodiment) of the top frame **30** (second top frame **32**), the impact of hitting the second top frame **32** is absorbed by the cover **33**.

(20) Next, the detailed configuration of the electronic hi-hat **1** will be described with reference to FIG. 2 and FIG. 3. (a) of FIG. 2 is a partially enlarged cross-sectional view of the electronic hi-hat **1** in which the IIa portion of FIG. 1 is enlarged, and (b) of FIG. 2 is a partially enlarged cross-sectional view of the electronic hi-hat **1** showing a state where the top cymbal **3** is closed from the state of (a) of FIG. 2. (a) of FIG. 3 is a partially enlarged cross-sectional view of the electronic hi-hat **1** showing a state where the top cymbal **3** is hit from the state of (b) of FIG. 2, and (b) of FIG. 3 is a partially enlarged cross-sectional view of the electronic hi-hat **1** showing a state where the bottom cymbal **2** and the top cymbal **3** are displaced beyond an initial position from the state of (a) of FIG. 3.

(21) As shown in FIG. 2, in a state where the pedal (not shown) that moves the top cymbal **3** up and down is released from depression (the state of (a) of FIG. 2), the top cymbal **3** is separated from the bottom cymbal **2**, and this state is an open state. When the pedal is depressed from the open state, the top cymbal **3** displaced downward comes into contact with the bottom cymbal **2** and becomes a closed state (the state of (b) of FIG. 2).

(22) The closed state of the top cymbal **3** caused by the depression of the pedal is detected by a displacement sensor (not shown) (which detects an upward/downward displacement amount of the top cymbal **3** caused by the upward/downward movement of the rod **6**), and this detection signal is also output to the sound source device (not shown). As a result, a musical tone corresponding to the performance that closes the top cymbal **3** is generated.

(23) Further, as shown in FIG. 3, when the top cymbal **3** is hit in the closed state (or when the top cymbal **3** comes into contact with the bottom cymbal **2** due to the displacement at the time of hitting), the bottom cymbal **2** and the top cymbal **3** swing in a state of being in contact with each other. Then, this embodiment forms a configuration in which, by smoothing the swinging of the bottom cymbal **2** and the top cymbal **3** in this way, the feeling at the time of hitting can be improved. This configuration will be described hereinafter.

(24) As shown in FIG. 2, the bottom frame **20** of the electronic hi-hat **1** includes a bow portion **20b** that gently slopes upward from the inner edge side to the outer edge side, an edge portion **20c** that extends outward in the radial direction from the outer edge of the bow portion **20b**. Each of the bow portion **20b** and the edge portion **20c** is formed continuously and integrally in the circumferential direction.

(25) An annular cushion material **21** is adhered to the upper surface of the edge portion **20c**, and an annular resin film **22** is adhered to the upper surface of the cushion material **21**. The cushion material **21** is formed using foamed resin, and the resin film **22** is formed using a material having a lower friction coefficient than the cushion material **21** (cloth made of polyester in this embodiment).

(26) The cushion material **21** and the resin film **22** are used to mitigate the impact at the time of contact (collision) between the bottom cymbal **2** and the top cymbal **3** as shown in FIG. 2, and smooth the sliding between the bottom cymbal **2** and the top cymbal **3** during swinging as shown in FIG. 3. In other words, a lightweight and flat plate shape outer edge portion of the bottom cymbal **2** and an outer edge portion of the top cymbal **3** constitute a smooth contact structure.

(27) From the lower surface of the second top frame **32** of the top cymbal **3**, a protrusion **32b** protrudes downward toward the resin film **22**, and when the top cymbal **3** is displaced, the protrusion **32b** comes into contact (collides) with the edge portion **20c** of the bottom frame **20** via the cushion material **21** and the resin film **22**.

(28) Because the edge portion **20c** is formed in a flat plate shape, the outer edge portion of the bottom frame **20** is relatively lightweight. As a result, even if the hit top cymbal **3** comes into contact with the bottom frame **20**, the weight of the bottom frame **20** is less likely to hinder the swinging (displacement caused by the hit) of the top cymbal **3**. Since the feeling at the time of

hitting the top cymbal **3** can be lightened, the feeling of hitting can be improved.

(29) The “flat plate shape” means that both the upper and lower surfaces of the edge portion **20c** are parallel, that is, the thickness (vertical dimension in FIG. 2) is constant from the inner edge to the outer edge of the edge portion **20c**. Further, “constant” means that the minimum and maximum thicknesses of the edge portion **20c** are within the range of $\pm 30\%$ of the average value of the thickness from the inner edge to the outer edge of the edge portion **20c**. Thus, by not forming unevenness (voids) such as ribs on the upper surface of the edge portion **20c** (opposing surface facing the top cymbal **3**), the flatness of the upper surface of the edge portion **20c**, that is, the flatness of the resin film **22** can be ensured. Therefore, the resin film **22** and the protrusion **32b** as shown in FIG. 3 can slide smoothly.

(30) In this embodiment, the thickness of the edge portion **20c** is set to 3 mm, but the thickness of the edge portion **20c** is preferably set to 0.5 times or more and 3 times or less of the thickness of the bow portion **20b** of the bottom frame **20** (3 mm in this embodiment), and more preferably formed with a thickness of 2 mm or more and 4 mm or less. As a result, it is possible to reduce the weight of the edge portion **20c** and improve the feeling of hitting while suppressing a decrease (damage) in the strength of the edge portion **20c**.

(31) Here, the conventional electronic hi-hat (for example, Japanese Laid-Open No. 2005-208555) has a configuration that a tube made of resin is fitted into the lower surface of the cover that covers the top frame, and the tube is brought into contact with the bottom cymbal when the top cymbal is displaced. In the case of such a configuration, in addition to that the number of parts increases, the tube may not properly contact the bottom cymbal due to elastic deformation of the cover.

(32) In contrast thereto, in this embodiment, the protrusion **32b** integrally formed with the second top frame **32** protrudes downward with respect to the lowermost surface of the cover **33**, and the protrusion **32b** is in contact with the edge portion **20c** of the bottom frame **20** (via the cushion material **21** and the resin film **22**). As a result, the number of parts of the electronic hi-hat **1** can be reduced. Further, unlike the configuration in which the tube is fitted into the cover **33**, the protrusion **32b** is not affected by the elastic deformation of the cover **33**. Therefore, the protrusion **32b** can be properly brought into contact with the resin film **22**, so the sliding between the protrusion **32b** and the resin film **22** can be effectively performed.

(33) On the other hand, in the case where the protrusion **32b** is integrally formed with the second top frame **32**, since the rigidity of the protrusion **32b** is relatively high, for example, when the protrusion **32b** directly comes into contact (collides) with the edge portion **20c**, the impact is large. Therefore, there arise the problems that the collision sound at the time of such contact is loud and the flat plate-shaped edge portion **20c** is easily damaged. In addition, when the bottom cymbal **2** and the top cymbal **3** slide as shown in FIG. 3, the protrusion **32b** and the edge portion **20c** are easily worn.

(34) In contrast thereto, in this embodiment, the cushion material **21** softer than the edge portion **20c** is laminated on the edge portion **20c**, and the resin film **22** having a lower friction coefficient than the cushion material **21** is laminated on the cushion material **21**. As a result, the impact at the time of contact (collision) between the bottom cymbal **2** and the top cymbal **3** as shown in FIG. 2 can be mitigated by the cushion material **21**, so the collision sound at the time of such contact can be reduced, and even when the edge portion **20c** is formed in a flat plate shape, damage to the edge portion **20c** can be suppressed. Further, when the bottom cymbal **2** and the top cymbal **3** slide as shown in FIG. 3, since the sliding can be smoothed by the resin film **22**, it is possible to prevent the protrusion **32b** and the edge portion **20c** from being worn.

(35) That is, the configuration in which the cushion material **21** and the resin film **22** are provided at the contact portion between the bottom frame **20** and the second top frame **32** is particularly effective when the protrusion **32b** integrally formed with the second top frame **32** is brought into contact with the flat plate-shaped edge portion **20c**.

(36) The second top frame **32** includes an outer peripheral portion **32c** that extends to the outer

peripheral side (right side in FIG. 2) with respect to the protrusion **32b**, a bent portion **32d** that bends downward from the outer edge of the outer peripheral portion **32c**, and an overhanging portion **32e** that projects from the lower end of the bent portion **32d** to the outer peripheral side. Each of these parts constitutes the edge (outer edge) portion of the second top frame **32**. Each of the outer peripheral portion **32c**, the bent portion **32d**, and the overhanging portion **32e** is formed continuously and integrally in the circumferential direction.

(37) The annular pressure-sensitive sensor **S2** is attached to the upper surface of the overhanging portion **32e**, and the pressure-sensitive sensor **S2** is covered with the cover **33**. When the edge portion of the second top frame **32** (cover **33**) is hit, the pressure-sensitive sensor **S2** is pressed by the cover **33**. As a result, the hit on the edge portion of the top cymbal **3** is detected.

(38) The outer edge portion of the cover **33** is folded downward so as to cover the outer peripheral surface and the lower surface of the overhanging portion **32e**, and the inner edge of the folded portion is inserted between the protrusion **32b** and the bent portion **32d** of the second top frame **32**. As a result, in addition to the function of smoothing the sliding at the time of contact with the bottom cymbal **2**, the protrusion **32b** can have a function of fixing the outer edge portion of the cover **33**.

(39) Here, as described above, the protrusion **32b** of the second top frame **32** protrudes downward with respect to the lowermost surface of the cover **33**, and in the state where the top cymbal **3** is closed, the protrusion **32b** is in contact with the resin film **22**. This contact position is defined as an initial position **P1** (see (b) of FIG. 2). As shown in FIG. 3, when the top cymbal **3** hit in the closed state tilts downward together with the bottom cymbal **2**, the contact position of the protrusion **32b** slides to a portion **P2** on the inner edge side of the resin film **22** with respect to the initial position **P1**. This is because the swing fulcrum of the bottom cymbal **2** and the swing fulcrum of the top cymbal **3** are vertically separated from each other.

(40) In addition, depending on the striking power to the top cymbal **3**, when the bottom cymbal **2** and the top cymbal **3** tilting downward return to the initial state, they may tilt upward beyond the initial state as shown in (b) of FIG. 3. At this time, the contact position of the protrusion **32b** slides to a portion **P3** on the outer edge side of the resin film **22** with respect to the initial position **P1**.

(41) That is, when the top cymbal **3** swings during hitting in the closed state, the protrusion **32b** comes into contact with the bottom cymbal **2** (resin film **22**) in a contact range **R** (predetermined range) from the portion **P2** on the inner edge side to the portion **P3** on the outer edge side of the bottom cymbal **2** (resin film **22**). Then, the edge portion **20c** is formed in a flat plate shape over a range wider than the contact range **R**. Thus, the edge portion **20c** is formed in a flat plate shape over the entire contact range **R** where the top cymbal **3** (protrusion **32b**) can come into contact to reduce the weight, so that the feeling at the time of hitting the top cymbal **3** can be effectively improved.

(42) Further, in order to smooth (stabilize) the sliding between the resin film **22** and the protrusion **32b** as shown in FIG. 3, it is preferable to make the tip of the protrusion **32b** constantly in contact with the resin film **22**. Therefore, in this embodiment, each of the edge portion **20c**, the cushion material **21**, and the resin film **22** is formed in a flat plate shape (the resin film **22** has a film shape) extending in a direction orthogonal to the protruding direction of the protrusion **32b**. As a result, the tip of the protrusion **32b** can be constantly slid on the upper surface of the resin film **22**, so that the sliding can be smoothed.

(43) An inner protrusion **20d** protrudes upward from the upper surface of the bottom frame **20** on the inner peripheral side with respect to the edge portion **20c**, and the inner protrusion **20d** is formed in an annular shape that is continuous over the entire circumference of the bottom frame **20**. As a result, the vicinity of the edge portion **20c** can be reinforced by the inner protrusion **20d**. Further, by forming the inner protrusion **20d** in the vicinity of the edge portion **20c**, it is possible to suppress the edge portion **20c** from being distorted due to heat shrinkage when the bottom frame **20** made of resin is molded. By suppressing the distortion of the edge portion **20c**, the flatness of the

upper surface of the edge portion **20c**, that is, the flatness of the resin film **22** can be ensured, so that the sliding between the resin film **22** and the protrusion **32b** as shown in FIG. **3** can be smoothed.

(44) An outer protrusion **20e** protrudes upward from the outer edge of the edge portion **20c**, and the outer protrusion **20e** is formed in an annular shape along the outer peripheral surfaces of the cushion material **21** and the resin film **22**. As a result, the outer peripheral surfaces of the cushion material **21** and the resin film **22** can be covered with the outer protrusion **20e**, so that the design of the electronic hi-hat **1** can be improved.

(45) Further, by forming the outer protrusion **20e** on the outer edge portion of the edge portion **20c**, the edge portion **20c** can be surrounded by the inner protrusion **20d** and the outer protrusion **20e** over the entire circumference in the circumferential direction. As a result, it is possible to more effectively suppress the distortion in the edge portion **20c** due to the above-mentioned heat shrinkage.

(46) As described above, in order to suppress the distortion in the edge portion **20c**, it is preferable to form each of the inner protrusion **20d** and the outer protrusion **20e** in an annular shape that is continuous in the circumferential direction. On the other hand, if the inner protrusion **20d** and the outer protrusion **20e** are formed in an annular shape that is continuous (complete without a gap) in the circumferential direction, when the bottom frame **20** is distorted (bent) by the load applied at the time of hitting, etc., there is no escape field for stress and the generated distortion cannot be absorbed, and as a result, the stress may be concentrated on the tips of the inner protrusion **20d** and the outer protrusion **20e** and cause cracks.

(47) In contrast thereto, in this embodiment, the bottom frame **20** is formed using fiber reinforced resin (glass fiber reinforced resin in this embodiment). As a result, even when the inner protrusion **20d** and the outer protrusion **20e** are formed in an annular shape, it is possible to suppress cracking of the inner protrusion **20d** and the outer protrusion **20e** due to the distortion (bending) of the bottom frame **20**.

(48) Further, since the bottom frame **20** is formed using fiber reinforced resin, the strength of the edge portion **20c** can also be secured. As a result, even when the edge portion **20c** is formed in a flat plate shape (thin), it is possible to prevent the edge portion **20c** from being damaged by contact (collision) with the top cymbal **3**. In other words, by forming the bottom frame **20** using fiber reinforced resin, as in the conventional electronic hi-hat (for example, Japanese Laid-Open No. 2005-208555), the edge portion of the bottom frame **20** can be formed in a flat plate shape (thin) without using a metal plate in the portion in contact with the top cymbal **3** or reinforcing with ribs or the like. Therefore, the number of parts can be reduced to reduce the product cost of the electronic hi-hat **1**, and the weight increase of the bottom frame **20** can be suppressed to improve the feeling at the time of hitting.

(49) Although the disclosure has been described based on the above embodiments, the disclosure is not limited to the above embodiments, and it can be easily inferred that various improvements and modifications can be made without departing from the gist of the disclosure.

(50) The above embodiment illustrates that the portion of the bottom frame **20** located in the contact range **R** with the top cymbal **3** is configured as the flat plate-shaped edge portion **20c**, but the disclosure is not necessarily limited thereto. For example, the bottom frame **20** may be formed in a flat plate shape in a range narrower than the contact range **R**.

(51) The above embodiment illustrates that the cushion material **21** and the resin film **22** are laminated on the edge portion **20c**, but the disclosure is not necessarily limited thereto. For example, in addition to the cushion material **21** and the resin film **22**, other members such as a metal plate may be attached to the edge portion **20c**. Further, the cushion material **21** and the resin film **22** may be omitted, and the edge portion **20c** may be brought into direct contact with the top frame **30** (second top frame **32**).

(52) The above embodiment illustrates that the protrusion **32b** protruding toward the edge portion

20c is formed integrally with the top frame **30** (second top frame **32**), but the disclosure is not necessarily limited thereto. For example, as in the conventional electronic hi-hat, a tube made of resin may be fitted into the lower surface of the top frame **30** (second top frame **32**) or the cover **33**, and the tube may be brought into contact with the bottom frame **20**. Further, instead of bringing the protrusion **32b** into contact with the bottom frame **20**, the top frame **30** (second top frame **32**) and the bottom frame **20** may be brought into contact with each other at the surfaces.

(53) The above embodiment illustrates that the top frame **30** includes the first top frame **31** and the second top frame **32**, but the disclosure is not necessarily limited thereto. For example, the top frame **30** may be composed of one or three or more frames.

(54) The above embodiment illustrates that the inner protrusion **20d** is provided on the inner peripheral side of the edge portion **20c** of the bottom frame **20** and the outer protrusion **20e** is provided on the outer peripheral side of the edge portion **20c**, but the disclosure is not necessarily limited thereto. For example, either one or both of the inner protrusion **20d** and the outer protrusion **20e** may be omitted.

(55) The above embodiment illustrates that the inner protrusion **20d** and the outer protrusion **20e** are formed in an annular shape (formed continuously in the circumferential direction), but the disclosure is not necessarily limited thereto. For example, the inner protrusion **20d** or the outer protrusion **20e** may be formed intermittently in the circumferential direction of the bottom frame **20** (for example, the inner protrusion **20d** extending in the radial direction is formed radially). With this configuration, when the bottom frame **20** is distorted (bent) by the load at the time of hitting, an escape field for the stress can be generated (the generated distortion is absorbed). Since the stress can be prevented from being concentrated on the inner protrusion **20d** and the outer protrusion **20e**, cracking of the inner protrusion **20d** and the outer protrusion **20e** can be suppressed.

(56) The above embodiment illustrates that each frame of the bottom frame **20** and the top frame **30** (first top frame **31** and second top frame **32**) is formed using glass fiber reinforced resin, but the disclosure is not necessarily limited thereto. For example, each frame may be formed using fiber reinforced resin reinforced with other fibers such as carbon fibers or other synthetic resin (for example, ABS resin or super engineering plastic) containing no fiber. That is, the material constituting each frame is not limited to the above-mentioned forms.

(57) The above embodiment illustrates that the edge portion **20c**, the cushion material **21**, and the resin film **22** are formed in a flat plate shape extending in the direction orthogonal to the protruding direction of the protrusion **32b** of the top frame **30** (second top frame **32**), but the disclosure is not necessarily limited thereto. For example, the edge portion **20c**, the cushion material **21**, and/or the resin film **22** may be inclined with respect to the direction orthogonal to the protruding direction of the protrusion **32b**.

(58) The above embodiment illustrates that the entire bottom frame **20** is formed using fiber reinforced resin, but the disclosure is not necessarily limited thereto. For example, only the edge portion **20c** of the bottom frame **20** may be formed using fiber reinforced resin.

Claims

1. An electronic hi-hat having a radial direction and a circumferential direction, comprising: a bottom cymbal that is swingably attached to a first rod; and a top cymbal that is arranged on an upper side of the bottom cymbal and is swingably attached to a second rod which is relatively displaced up and down with respect to the first rod, wherein the bottom cymbal comprises a first frame that forms a skeleton of the bottom cymbal, and the first frame is formed continuously and integrally in the radial direction and the circumferential direction, the first frame comprises: a bow portion; a contact portion which is an edge portion formed in a flat plate shape over a contact range where the top cymbal comes into contact when the top cymbal is displaced; and an inner protrusion

that protrudes from a portion on an inner peripheral side with respect to the contact portion toward the top cymbal, wherein the contact portion continuously and integrally extends outward in the radial direction from the bow portion, and the bow portion and the contact portion are formed continuously and integrally in the circumferential direction of the first frame, wherein an upper surface and a lower surface of the contact portion are parallel, and a thickness of the contact portion in a vertical dimension is constant from an inner edge to an outer edge of the contact portion, wherein when the top cymbal comes into contact with the bottom cymbal due to a displacement at a time of hitting, the bottom cymbal and the top cymbal swing in a state of being in contact with each other, wherein when the electronic hi-hat is not hit, the flat plate shape of the contact portion is positioned in a direction substantially perpendicular to an axial direction of the first rod, wherein a bottom surface of the first frame has an angle between the contact portion and the bow portion, and the angle is positioned between the inner protrusion and the contact portion.

2. The electronic hi-hat according to claim 1, wherein a surface of the contact portion facing the top cymbal is formed in a flat shape having no unevenness.

3. The electronic hi-hat according to claim 1, wherein a range for contact with the top cymbal from an inner edge side to an outer edge side of the first frame when the top cymbal swings with respect to the second rod is configured as the contact portion having the flat plate shape.

4. The electronic hi-hat according to claim 1, wherein the bow portion constitutes a portion on an inner peripheral side with respect to the contact portion, and the contact portion is formed with a thickness that is 0.5 times or more and 3 times or less a thickness of the bow portion.

5. The electronic hi-hat according to claim 1, wherein the top cymbal comprises a second frame that forms a skeleton of the top cymbal, the second frame comprises a protrusion that protrudes toward the contact portion and is formed integrally with the second frame, and the protrusion and the contact portion come into contact with each other when the top cymbal is displaced.

6. The electronic hi-hat according to claim 5, wherein the bottom cymbal comprises: a cushioning material that is laminated on the contact portion and is softer than the contact portion, and a low friction material that is laminated on the cushioning material and has a lower friction coefficient than the cushioning material, wherein the protrusion and the contact portion come into contact with each other via the cushioning material and the low friction material.

7. The electronic hi-hat according to claim 6, wherein the contact portion, the cushioning material, and the low friction material are formed in the flat plate shape that extends in a direction orthogonal to a protruding direction of the protrusion.

8. The electronic hi-hat according to claim 1, wherein at least the contact portion of the first frame is formed using fiber reinforced resin.

9. An electronic hi-hat having a radial direction and a circumferential direction, comprising: a bottom cymbal, having a first frame that forms a skeleton of the bottom cymbal, and being swingably attached to a first rod, and the first frame is formed continuously and integrally in the radial direction and the circumferential direction; and a top cymbal, that is arranged on an upper side of the bottom cymbal and is swingably attached to a second rod which is relatively displaced up and down with respect to the first rod, wherein a lightweight outer edge portion of the bottom cymbal and an outer edge portion of the top cymbal constitute a smooth contact structure, the first frame comprises: a bow portion; a contact portion which is the lightweight outer edge portion formed in a flat plate shape over a contact range where the top cymbal comes into contact when the top cymbal is displaced; and an inner protrusion that protrudes from a portion on an inner peripheral side with respect to the contact portion toward the top cymbal, wherein the contact portion continuously and integrally extends outward in the radial direction from the bow portion, and the bow portion and the contact portion are formed continuously and integrally in the circumferential direction of the first frame, wherein an upper surface and a lower surface of the contact portion are parallel, and a thickness of the contact portion in a vertical dimension is constant from an inner edge to an outer edge of the contact portion, wherein when the top cymbal

comes into contact with the bottom cymbal due to a displacement at a time of hitting, the bottom cymbal and the top cymbal swing in a state of being in contact with each other, wherein when the electronic hi-hat is not hit, the flat plate shape of the contact portion is positioned in a direction substantially perpendicular to an axial direction of the first rod, wherein a bottom surface of the first frame has an angle between the contact portion and the bow portion, and the angle is positioned between the inner protrusion and the contact portion.

10. An operating method of an electronic hi-hat having a radial direction and a circumferential direction, which comprises a bottom cymbal that is swingably attached to a first rod, and a top cymbal that is arranged on an upper side of the bottom cymbal and is swingably attached to a second rod which is relatively displaced up and down with respect to the first rod, wherein the bottom cymbal comprises a first frame that forms a skeleton of the bottom cymbal, and the first frame is formed continuously and integrally in the radial direction and the circumferential direction, the first frame comprises: a bow portion; a contact portion which is an edge portion formed in a flat plate shape over a contact range where the top cymbal comes into contact when the top cymbal is displaced; and an inner protrusion that protrudes from a portion on an inner peripheral side with respect to the contact portion toward the top cymbal, wherein the contact portion continuously and integrally extends outward in the radial direction from the bow portion, and the bow portion and the contact portion are formed continuously and integrally in the circumferential direction of the first frame, wherein an upper surface and a lower surface of the contact portion are parallel, and a thickness of the contact portion in a vertical dimension is constant from an inner edge to an outer edge of the contact portion, wherein when the top cymbal comes into contact with the bottom cymbal due to a displacement at a time of hitting, the bottom cymbal and the top cymbal swing in a state of being in contact with each other, wherein when the electronic hi-hat is not hit, the flat plate shape of the contact portion is positioned in a direction substantially perpendicular to an axial direction of the first rod, wherein a bottom surface of the first frame has an angle between the contact portion and the bow portion, and the angle is positioned between the inner protrusion and the contact portion.

11. The operating method of an electronic hi-hat according to claim 10, wherein a surface of the contact portion facing the top cymbal is formed in a flat shape having no unevenness.

12. The operating method of an electronic hi-hat according to claim 10, wherein a range for contact with the top cymbal from an inner edge side to an outer edge side of the first frame when the top cymbal swings with respect to the second rod is configured as the contact portion having the flat plate shape.

13. The operating method of an electronic hi-hat according to claim 10, wherein the bow portion constitutes a portion on an inner peripheral side with respect to the contact portion, and the contact portion is formed with a thickness that is 0.5 times or more and 3 times or less a thickness of the bow portion.

14. The operating method of an electronic hi-hat according to claim 10, wherein the top cymbal comprises a second frame that forms a skeleton of the top cymbal, the second frame comprises a protrusion that protrudes toward the contact portion and is formed integrally with the second frame, and the protrusion and the contact portion come into contact with each other when the top cymbal is displaced.

15. The operating method of an electronic hi-hat according to claim 14, wherein the bottom cymbal comprises: a cushioning material that is laminated on the contact portion and is softer than the contact portion, and a low friction material that is laminated on the cushioning material and has a lower friction coefficient than the cushioning material, wherein the protrusion and the contact portion come into contact with each other via the cushioning material and the low friction material.

16. The operating method of an electronic hi-hat according to claim 15, wherein the contact portion, the cushioning material, and the low friction material are formed in the flat plate shape that extends in a direction orthogonal to a protruding direction of the protrusion.

17. The operating method of an electronic hi-hat according to claim 10, wherein the first frame comprises an outer protrusion that protrudes from a portion on an outer peripheral side with respect to the contact portion toward the top cymbal, and the contact portion is surrounded by the inner protrusion and the outer protrusion that extend over an entire circumference of the first frame.
 18. The operating method of an electronic hi-hat according to claim 10, wherein at least the contact portion of the first frame is formed using fiber reinforced resin.
 19. The electronic hi-hat according to claim 1, wherein the thickness of the contact portion in the vertical dimension is constant from the inner edge to the outer edge of the contact portion, wherein minimum and maximum of the thicknesses of the contact portion are within a range of $\pm 30\%$ of an average value of the thickness from the inner edge to the outer edge of the contact portion.
 20. The electronic hi-hat according to claim 1, wherein a bottom surface of first frame at the edge portion is substantially parallel with a contact surface opposite to the bottom surface of the first frame.
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